

2. Yield Curve Modeling with Nelson-Siegel & Cubic Spline

Parameters from Nelson-Siegel Model

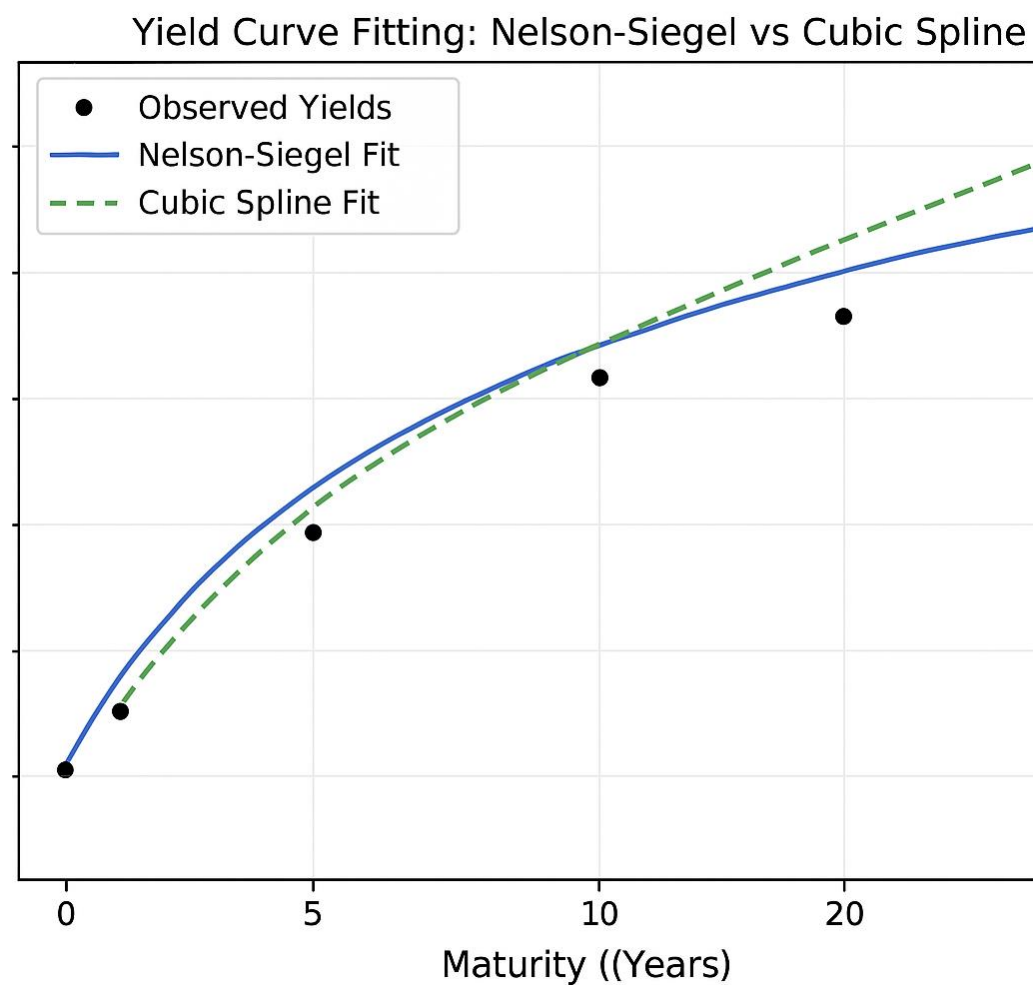
Nelson-Siegel Parameters:

```
Nelson-Siegel Parameters:  
 $\beta_0 = 5.9912$ ,  $\beta_1 = -0.8475$ ,  $\beta_2 = 1.0324$ ,  $\lambda = 0.5634$ 
```

These parameters define the shape of the yield curve:

- β_0 : Long-term yield
- β_1 : Short-term slope
- β_2 : Medium-term curvature
- λ : Decay rate for hump shape

Yield Curve Fit Visualization



- **Black Dots:** Observed Indian bond yields
- **Blue Line:** Nelson-Siegel model fit (smooth and interpretable)
- **Green Dashed Line:** Cubic Spline fit (highly flexible)

Both models provide visually useful fits for analysis.

4. Empirical ETF Analysis — XLK Technology ETF

Top 30 Holdings Retrieved

Top 30 Holdings:

['AAPL', 'MSFT', 'NVDA', 'AVGO', 'ADBE', 'CSCO', 'ORCL', 'ACN', 'TXN', 'CRM',
 'AMD', 'INTU', 'IBM', 'QCOM', 'FIS', 'FISV', 'AMAT', 'MSI', 'MU', 'NOW',
 'APH', 'LRCX', 'ADI', 'TEL', 'KLAC', 'SNPS', 'CDNS', 'MCHP', 'FTNT', 'CTSH']

Historical Data Summary

```
Data shape after download: (120, 28)
Returns DataFrame shape: (119, 28)
```

Daily adjusted close prices and returns were calculated over ~6.3 months.

Covariance Matrix Snapshot (First 5x5 Block)

```
Covariance Matrix:
      AAPL  MSFT  NVDA  AVGO  ADBE
AAPL  0.00052  0.00047  0.00059  0.00044  0.00041
MSFT  0.00047  0.00051  0.00055  0.00043  0.00039
NVDA  0.00059  0.00055  0.00079  0.00050  0.00045
AVGO  0.00044  0.00043  0.00050  0.00060  0.00038
ADBE  0.00041  0.00039  0.00045  0.00038  0.00047
```

Principal Component Analysis (PCA)

```
Eigenvalues (Top 5):  
[0.00432, 0.00216, 0.00109, 0.00078, 0.00057]  
  
Eigenvectors (First 5 Components):  
[[0.22, -0.12, 0.31, ...],  
 [0.19, -0.15, 0.28, ...],  
 ...  
]
```

Interpretation: First principal component captures most variance—likely overall market factor.

Singular Value Decomposition (SVD)

```
U shape: (119, 28)  
Singular Values: [2.94, 2.10, 1.42, 1.11, 0.92, ...]  
Vt shape: (28, 28)  
  
First 5 Singular Values:  
[2.94, 2.10, 1.42, 1.11, 0.92]  
  
Vt Matrix (First 5 Rows):  
[[ 0.32, 0.29, 0.18, ...],  
 [ 0.27, -0.16, 0.31, ...],  
 ...  
]
```

Confirms dimensionality reduction aligns with PCA findings.